Detailed proofs

This file includes the detailed proofs of Theorem 3.6 and Corollary 3.5 in the main text.

1 Detailed proof of Theorem 3.6

Theorem 3.6 (Optimality of the OPAD distribution in KL divergence). Let p be a discrete target distribution on support $\mathcal{X}$ with unnormalized version $\tilde{p}$. Let P be an arbitrary distribution with support $\mathcal{D} \subseteq \mathcal{X}$.

1. The greatest lower bound on the Kullback–Leibler divergence of P from the target, that is, $D_{\mathrm{KL}}(P \,||\, p)$, is the negative log of the probability mass that the target assigns to the support of P:

$$\inf_{P:\mathrm{supp}(P)=\mathcal{D}} D_{\mathrm{KL}}(P \,||\, p) \;=\; -\log\big(p(\mathcal{D})\big). \tag{1}$$

2. This minimum is achieved uniquely by the OPAD distribution on $\mathcal{D}$, i.e. if and only if

$$P(x) = \frac{\tilde{p}(x)}{\sum_{y\in\mathcal{D}} \tilde{p}(y)}, \qquad \forall x \in \mathcal{D}. \tag{2}$$

Proof. Let

$$\phi(u) := -\log u, \qquad g(x) := \frac{p(x)}{P(x)}, \qquad x \in \mathcal{D}.$$

Since ϕ is strictly convex on $(0, \infty)$, Jensen's inequality applies.

Because $\mathrm{supp}(P) = \mathcal{D}$, we have $P(x) > 0$ for all $x \in \mathcal{D}$, so g is well-defined on $\mathcal{D}$. Also, since $\mathcal{D} \subseteq \mathcal{X}$ and $\mathcal{X}$ is the support of p, we have $p(x) > 0$ for all $x \in \mathcal{D}$.

Now

$$\begin{aligned} D_{\mathrm{KL}}(P \,||\, p) &= \sum_{x\in\mathcal{D}} P(x) \log\left(\frac{P(x)}{p(x)}\right) \\ &= \sum_{x\in\mathcal{D}} P(x) \left(-\log\left(\frac{p(x)}{P(x)}\right)\right) \\ &= \sum_{x\in\mathcal{D}} \phi(g(x))\, P(x). \end{aligned} \tag{3}$$

Applying Jensen's inequality to the probability distribution P, the function g, and the strictly convex function ϕ, we obtain

$$\begin{aligned} D_{\mathrm{KL}}(P \,||\, p) &= \sum_{x\in\mathcal{D}} \phi(g(x))\, P(x) \\ &\geq \phi\left(\sum_{x\in\mathcal{D}} g(x)\, P(x)\right) \\ &= -\log\left(\sum_{x\in\mathcal{D}} \frac{p(x)}{P(x)}\, P(x)\right) \\ &= -\log\left(\sum_{x\in\mathcal{D}} p(x)\right) \\ &= -\log\big(p(\mathcal{D})\big). \end{aligned} \tag{4}$$

Since this lower bound holds for every probability distribution P with support $\mathcal{D}$, it follows that

$$\inf_{P:\mathrm{supp}(P)=\mathcal{D}} D_{\mathrm{KL}}(P \,||\, p) \geq -\log\big(p(\mathcal{D})\big).$$

We now characterise the equality case. Because ϕ is strictly convex, equality in Jensen's inequality holds if and only if $g(x)$ is constant on the support of P. Since the support of P is exactly $\mathcal{D}$, equality in (4) holds if and only if there exists a constant $c > 0$ such that

$$g(x) = \frac{p(x)}{P(x)} = c, \qquad \forall x \in \mathcal{D}.$$

Equivalently,

$$P(x) = \frac{p(x)}{c}, \qquad \forall x \in \mathcal{D}.$$

Since P is a probability distribution,

$$1 = \sum_{x\in\mathcal{D}} P(x) = \frac{1}{c} \sum_{x\in\mathcal{D}} p(x) = \frac{p(\mathcal{D})}{c},$$

and therefore

$$c = p(\mathcal{D}).$$

Substituting this back gives

$$P(x) = \frac{p(x)}{p(\mathcal{D})}, \qquad \forall x \in \mathcal{D}.$$

Hence the lower bound in (4) is attained, and so

$$\inf_{P:\mathrm{supp}(P)=\mathcal{D}} D_{\mathrm{KL}}(P \,||\, p) = -\log\big(p(\mathcal{D})\big),$$

which proves (1).

Finally, using

$$p(x) = \frac{\tilde{p}(x)}{\sum_{z\in\mathcal{X}} \tilde{p}(z)},$$

we obtain

$$P(x) = \frac{p(x)}{\sum_{y\in\mathcal{D}} p(y)} = \frac{\tilde{p}(x)}{\sum_{y\in\mathcal{D}} \tilde{p}(y)}, \qquad \forall x \in \mathcal{D},$$

which proves (2). The equality condition in Jensen's inequality also shows that this minimiser is unique. $\square$

2 Detailed proof of updated Theorem 3.4

Theorem 3.4 (Per-Chain Variance Reduction by OPAD Mapping) Let $\{X_n\}_{n=1}^N$ be a stationary, reversible Markov chain on a discrete space $\mathcal{X}$ with stationary distribution $p(x)$. The variance of the OPAD estimator of the target quantity is less than or equal to that of the corresponding MC estimator:

$$\mathrm{Var}[\hat{\theta}_{\text{OPAD}}] \leq \mathrm{Var}[\hat{\theta}_{\text{MC}}].$$

Proof. Let $\mathcal{D}$ denote the set of distinct visited states. For a reversible Markov chain at stationarity, the conditional expectation of the fraction of samples in each state is given by the renormalized stationary probabilities:

$$\mathbb{E}\Big[\frac{N_x}{N}\Big|\mathcal{D}\Big] = \frac{p(x)}{\sum_{y\in\mathcal{D}} p(y)} = P_{\mathcal{D}}(x).$$

Therefore, the OPAD estimator is the conditional expectation of the MC estimator with respect to the set of visited states $\mathcal{D}$:

$$\mathbb{E}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}] = \sum_{x\in\mathcal{D}} \mathbb{E}\Big[\frac{N_x}{N}\Big|\mathcal{D}\Big] f(x) = \sum_{x\in\mathcal{D}} P_{\mathcal{D}}(x) f(x) = \hat{\theta}_{\text{OPAD}}.$$

Applying the law of total variance to $\hat{\theta}_{\text{MC}}$ gives

$$\mathrm{Var}[\hat{\theta}_{\text{MC}}] = \mathbb{E}[\mathrm{Var}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]] + \mathrm{Var}[\mathbb{E}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]].$$

Substituting $\mathbb{E}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}] = \hat{\theta}_{\text{OPAD}}$, we have

$$\mathrm{Var}[\hat{\theta}_{\text{MC}}] = \mathbb{E}[\mathrm{Var}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]] + \mathrm{Var}[\hat{\theta}_{\text{OPAD}}].$$

Since the conditional variance $\mathrm{Var}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]$ is inherently non-negative, its expectation satisfies $\mathbb{E}[\mathrm{Var}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]] \geq 0$. We thus obtain the inequality

$$\mathrm{Var}[\hat{\theta}_{\text{OPAD}}] \leq \mathrm{Var}[\hat{\theta}_{\text{MC}}].$$

$\square$

3 Detailed proof of Corollary 3.5 in the main text

Corollary 3.5 (Between-chain variance reduction of OPAD estimators). Let $C_1, \ldots, C_K$ be $K \geq 2$ independent Markov chains, each of length N, from a discrete target distribution $p(x)$ on support $\mathcal{X}$, generated under the stationary and reversible conditions specified in Theorem 3.4. For each chain C_i, let $\hat{\theta}^{(i)}_{\text{MC}}$ denote the standard Monte Carlo estimator and let $\hat{\theta}^{(i)}_{\text{OPAD}}$ denote the corresponding OPAD estimator.

For $T \in \{\text{MC}, \text{OPAD}\}$, define the between-chain sample variance by

$$S_T^2 := \frac{1}{K-1}\sum_{i=1}^{K}\Big(\hat{\theta}_T^{(i)} - \bar{\theta}_T\Big)^2, \qquad \bar{\theta}_T := \frac{1}{K}\sum_{i=1}^{K}\hat{\theta}_T^{(i)}.$$

Then

$$\mathbb{E}[S^2_{\text{OPAD}}] \ \leq \ \mathbb{E}[S^2_{\text{MC}}].$$

Proof. Fix $T \in \{\text{MC}, \text{OPAD}\}$. Since the chains $C_1, \ldots, C_K$ are independent and identically generated (each starting from or having reached stationarity), the estimators

$$\hat{\theta}_T^{(1)}, \ldots, \hat{\theta}_T^{(K)}$$

are i.i.d. random variables. Therefore, S_T^2 is the unbiased sample variance computed from K i.i.d. draws, and its expectation equals the common variance of those observations:

$$\mathbb{E}[S_T^2] = \mathrm{Var}\Big(\hat{\theta}_T^{(i)}\Big) \qquad \forall i \in \{1, \ldots, K\}.$$

Applying this identity with $T = \textsc{mc}$ and $T = \textsc{opad}$, and invoking the variance reduction established in Theorem 3.4, we obtain:

$$\mathbb{E}[S^2_{\textsc{opad}}] = \mathrm{Var}\Big(\hat{\theta}^{(i)}_{\textsc{opad}}\Big) \leq \mathrm{Var}\Big(\hat{\theta}^{(i)}_{\textsc{mc}}\Big) = \mathbb{E}[S^2_{\textsc{mc}}] \qquad \forall i \in \{1, \ldots, K\},$$

which proves the claim. $\square$